



Final Assignment

Poverty Prediction Challenge

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1 Introduction

This report analyzing the solution of [Poverty Prediction Challenge](#). This challenge is focusing on accurately find the poverty measurement worldwide. It simulates a common real-word scenario faced by World Bank economists which produce the dataset used in this problem. The goal of this project is to develop models which predict both poverty rates and per capita household consumption from anonymized historical survey data.

2 Methodology

This problem is complicated and for that reason it is necessary to analyze not only the data by itself but also understand the meaning of the values problem focuses on. Problem provides not only the dataset by itself but also description of each feature to get a better view of the data. Training dataset consists of 104235 rows and 88 columns (features). Some features are indexes which are not providing any insides of the data but on the other hand there are important features which contributes to good training results. In the section below there are most of the features separated in categories followed with a small description. This will help to understand the importance of each feature and also handle them if there are missing values or imbalance anomalies.

2.1 Variable Categorization – Household Welfare Dataset

1) Identification & Survey Design Variables

(Not used as predictors, required for survey design and weighting)

VARIABLE	DESCRIPTION
SAMPLE	Data sample type
HHID	Household unique identifier
COM	Household member identifier
WEIGHT	Household sampling weight
STRATA	Stratification variable

Table 1: Identification & Survey Design Variables

2) Target Variable (Outcome)

VARIABLE	DESCRIPTION
CONS_PPP17	Per-capita daily household expenditure (PPP 2017)

Table 2: Target Variable (Outcome)

3) Utility Expenditures

VARIABLE	DESCRIPTION
UTL_EXP_PPP17	Household expenditure on utilities (PPP 2017)

Table 3: Utility Expenditures

4) Household Demographics

VARIABLE	DESCRIPTION
MALE	Household head is male
AGE	Age of household head
HSIZE	Household size
NUM_CHILDREN5	Number of children under 5
NUM_CHILDREN10	Number of children aged 5–10
NUM_CHILDREN18	Number of children aged 10–18
NUM_ADULT_FEMALE	Number of female adults (18–69)
NUM_ADULT_MALE	Number of male adults (18–69)
NUM_ELDERLY	Number of elderly (70+)

Table 4: Household Demographics

5) Housing & Infrastructure Characteristics

VARIABLE	DESCRIPTION
OWNER	Dwelling ownership
DWELTYP	Type of dwelling
WATER	Access to water supply
WATER_SOURCE	Main drinking water source
TOILET	Presence of toilet facilities
SEWER	Toilet connected to sewer
SANITATION_SOURCE	Main sanitation system
ELECT	Access to electricity

Table 5: Housing & Infrastructure Characteristics

6) Employment & Labor Market

VARIABLE	DESCRIPTION
EMPLOYED	Household head is employed
SECTOR1D	Employment sector of household head (ISIC 1-digit)
ANY_NONAGRIC	Any household member in non-agricultural work
SWORKERSHH	Share of adults working in the household
SFWORKERSHH	Share of adults in formal employment

Table 6: Employment & Labor Market

7) Education

VARIABLE	DESCRIPTION
EDUC_MAX	Highest education level in household
SHARE_SECONDARY	Share of adults with secondary education

Table 7: Education

8) Geography & Urbanization

VARIABLE	DESCRIPTION
URBAN	Urban residence indicator
REGION1	Region 1
REGION2	Region 2
REGION3	Region 3
REGION4	Region 4
REGION5	Region 5
REGION6	Region 6
REGION7	Region 7

Table 8: Geography & Urbanization

There is also an important category which is focusing on food consumption indicators. This category is labeled and each label means is about a different type of food. There are 8 labels:

- 1) Cereals & Grains
- 2) Dairy & Eggs
- 3) Meat, Fish & Animal Products
- 4) Vegetables & Legumes
- 5) Fruits
- 6) Fats, Condiments & Spices
- 7) Beverages & Prepared Items
- 8) Other Food Consumption

These labels are custom and there are not mentioned in the real dataset in that way

This work is essential so user can understand the data before use code to get more insights.

2.2 Exploratory Data Analysis

This section will focus on more advanced techniques for Exploratory Data Analysis (EDA). An external library was used, since there are many features to be analyzed. This library is called `ydata_profiling` and it is a leading package for data profiling, that automates and standardizes the generation of detailed reports. This report helps to get the distribution of continuous prices, find imbalanced or columns with many zeros and the correlations between the data by providing a heatmap and a helpful table of feature information. Also, it is easier to see the type of the data and the corresponding analysis depending on the type, for example if they are numbers, you can see their distribution while if they are words, you can see their frequency (with a word of cloud)

First, the missing values problem appears in a few features (Figure 1). There are also more features with missing values but in these columns, the percentage is high enough to be mentioned.

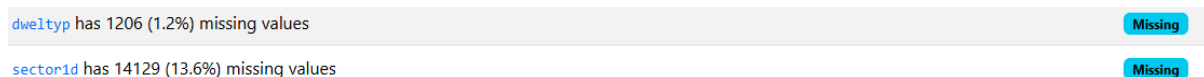


Figure 1: Missing values for `dweltyp` and `sector1d` (Screenshot from Ydata profile)

Before encountering this problem, it is important to plot all missing values to find a way to solve them separately or all together with the same technique. It is clear that many other features had nan values inside (Figure 2). Most of them come from consumed category and a few from `utl_exp_ppp17`, `employed` and `share_secondary`.

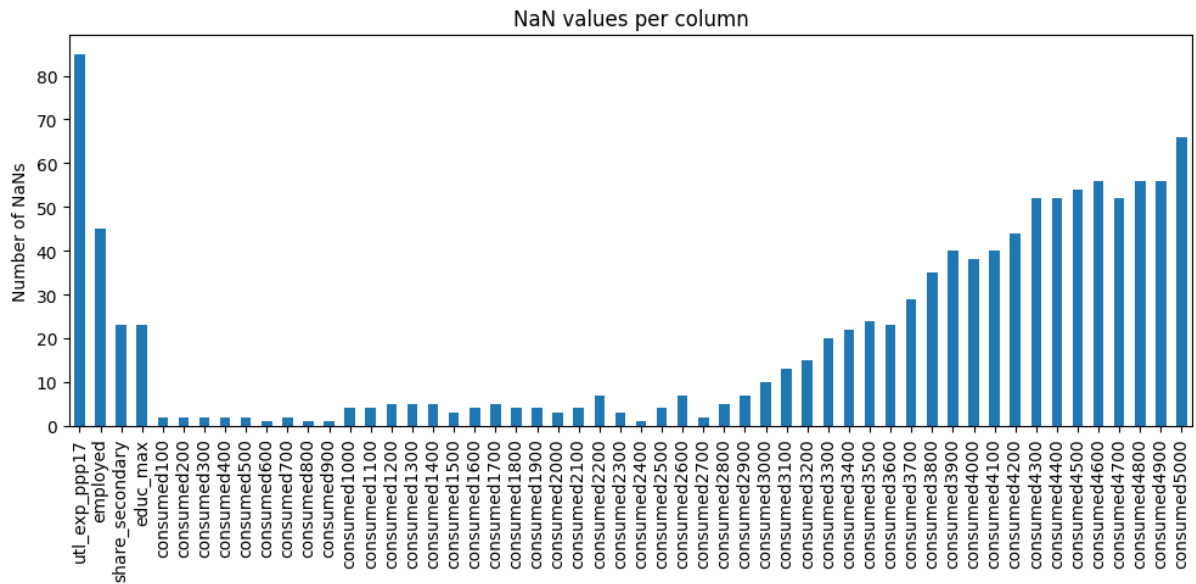


Figure 2: Bar plot counting low percentage nan values

It was a bit challenging to handle this situation, since consume features might miss because of the people who get the recordings but on the other hand utl_exp_ppp17, employed and share_secondary might miss because of a different reason. It was way safer to replace these nan values with 'unknown' word to find if there is a pattern in these families which are missing this information. This technique applied and in dweltype and sector1d. Since it is important to clear the dataset from nan values, drop them (there are only from consumed features).

At this point, necessary to note, that nan values problem was encounter with different technique in test set than training set. In test set it is not allowed to drop columns and rows since final submit will be invalid. The problem in the previous technique is that nan consume features drop from the dataset. Also, in the test set there are nan values in the same columns. To encounter this problem a model was trained to fill missing values. To ensure speed and accuracy XGBoost model was selected. With this algorithm consume values were filled and for continuous values nan values were filled with the mean value of the column.

Now dataset is clear from nan values and next step is to get advantage of the report and find the correlation between features. Heatmap is the easiest visualization technique to get a summarization of this (Figure 4). It is not easy to be read but there are many blue boxes and many dark blue boxes meaning that there is a high correlation between many columns. Cons_ppp17 is the target value and it is important that it has high correlation with strata and utl_exp_ppp17.

cons_ppp17 is highly overall correlated with strata and 1 other fields strata, utl_exp_ppp17 High correlation

Figure 3: High correleation features with target value (Screenshot from Ydata profile)

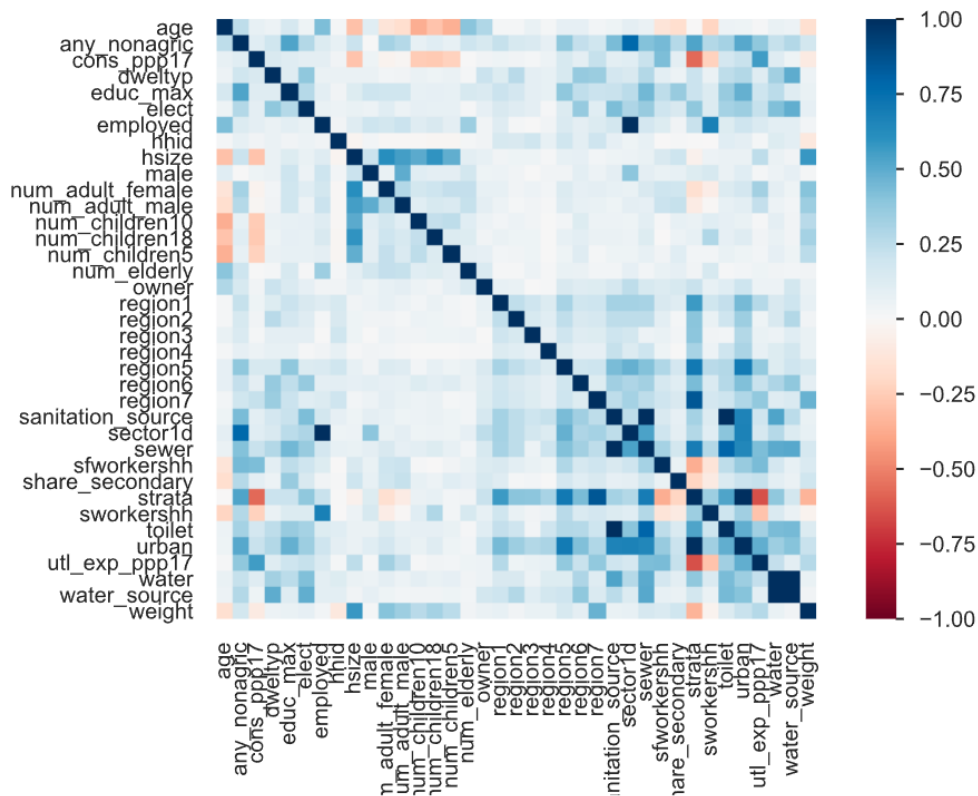


Figure 4: Histogram of features

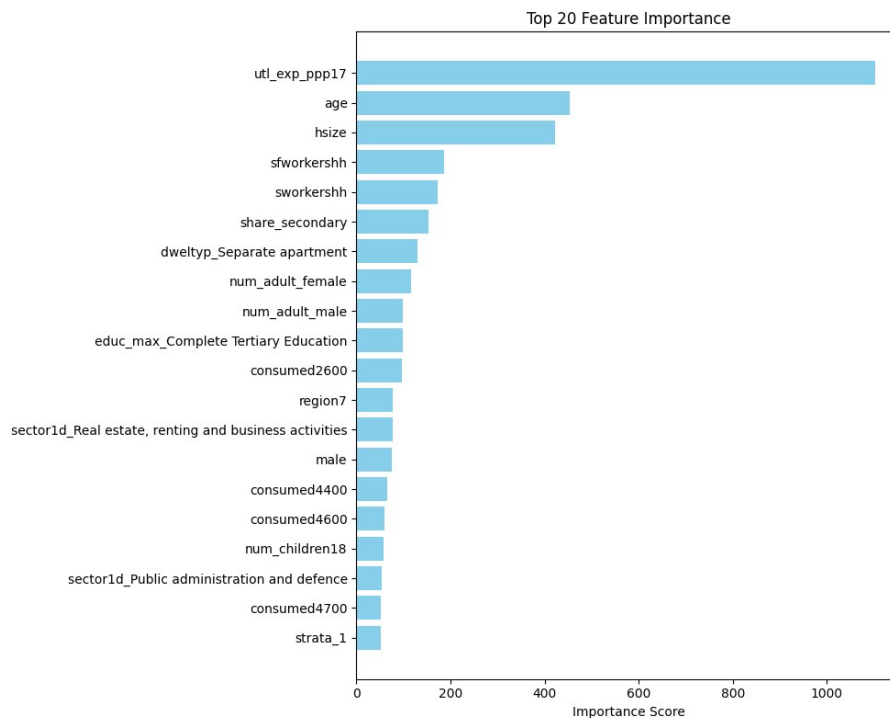


Figure 5: XGBoost feature importance bar plot

Report also provides many features which needs Label encoding since they have 2 labels, but they are not numerical (Figure 6).

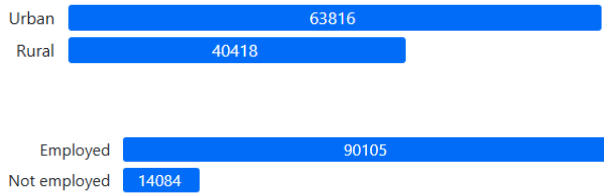


Figure 6: Categorical features with non-numerical values (Screenshot from Ydata profile)

Since this phenomenon appears in a big range of features, a better approach was to find columns with two unique values and map the word with 1 (True) or 0 (False).

That was not the only technique used to clear the data. There are also many more which has more than one labels numerical or not (Figure 7)

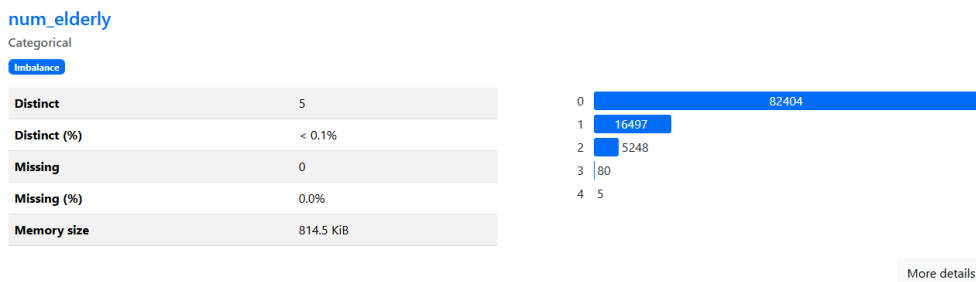


Figure 7: details of number of elder people in a family (Screenshot from Ydata profile)

In these situations, it is necessary to understand if these words or numbers have a meaning, if 1 is greater than 2 or reverse and if 'A piped sewer system' is a greater sanitation_source than 'A septic tank'. If first suspicion was correct, then data converted into numerical with 1 be the least important but if it is not then one hot encoding technique was applied to separate the column since data has no order.

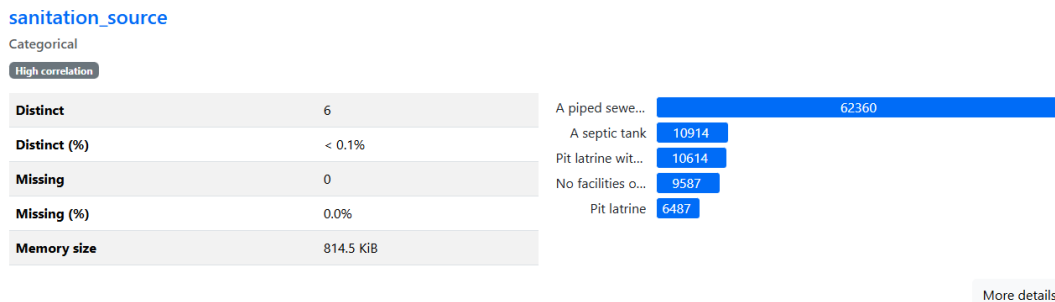


Figure 8: Details of sanitation source feature (Screenshot from Ydata profile)

Also, it is important to mention that features didn't have outliers, and enough features didn't have high imbalance.

After applying these filters dataset features change from 88 to 134. It is important to apply the same techniques in both train and test set to ensure that column size will match and the data type and range be the same in both sets.

2.3 Machine Learning Regression Models

Data are ready to be fed into 3 different regression models and one Deep Neural Network model. Models were chosen to find not only the best but also to create a baseline and a better understanding of the problem.

Before applying all data (train and test set), a better approach was to split only the training set into training and test. This method will help to have labels in test set, and this is very helpful to track model performance using metrics such as RMSE and R^2 . Since dataset has more than 100k rows, train set had the 90% of the data while test set only 10% of it.

First regression model is used as a baseline. Linear Regression is very efficient and light. It is easy to be trained and provides a good perspective of the dataset. After training, Linear Regression had good results for such a simple model (Table 9).

METRICS	VALUES
RMSE	6.422
R^2	0.528

Table 9: Linear regression model training results

The RMSE indicates that the deviate from the true values is approximately 6.4 PPP-adjust units. This level of error suggests a moderate but non-negligible prediction uncertainty.

The R^2 value is 0.53 shows that the model approximately 53% of the variance can be explained across households.

Overall, these results are consistent with expectations for a linear baseline model applied to complex household survey data. The model captures broad linear trends; its performance suggests that more flexible nonlinear models are required to better represent the complexity of household consumption patterns.

Next model is an SVR model which had some limitations. This model is heavy to be trained, and it needs more than 4 hours for a nonlinear kernel. This problem led to create only one version of it since it was difficult to tune the model by trying different parameter values. Support Vector Regression parameters are the default with the only thing change is the gamma which is set to 'auto' to fit better into data rather than adding hardcoded a value which might be catastrophic for the model. The model didn't have great performance regarding Linear Regression results (Table 10)

METRICS	VALUES
RMSE	7.38
R^2	0.37
MAE	4.605

Table 10: SVR model training results

SVR models achieve **RMSE = 7.38**, **MAE = 4.61** and **$R^2 = 0.38$** in test set. The high RMSE indicates model is sensitive to large outliers while MAE demonstrates that the errors are systematically larger and not just the result of a few outliers.

Value of R^2 indicates that 38% can be explained of the variation in consumption, which shows a limit in the ability to capture the relationship between the characteristics and the target variable.

Overall, without any hyperparameter tuning SVR does not perform well on household survey data with high heterogeneity and nonlinear relationships.

The last Regression model is XGBoost. After analysing the results of Linear Regression and SVR it is important to get into an advanced and efficient algorithm. SVR is also advanced but very slow regarding the speed and results of XGBoost. This model provides better performance than the other 2 models (Table 11).

METRICES	VALUES
RMSE	5.568
R ²	0.645
MAE	3.096

Table 11: XGBoost model training results

RMSE decrease, indicates that predictions are closer to the mean of per capita daily consumption. Also, R² achieve 64.5% score which is significantly higher than other models, meaning that XGBoost adapt better to the dataset. MAE value is also acceptable, something which is particularly important for the reliable estimation of consumption and poverty rates.

Overall, XGBoost model as a nonlinear tree boosting model, is better to capture more complex relationships between household characteristics and consumption.

In every model weight removed from feature and added to model.fit as (sample_weight=np.array(test_weight))

2.4 Deep Neural Network model

After reviewing the model performances, next step is to create a deep neural network model to compare the performance between them. To create the architecture of a neural network is a difficult task since there are many parameters and methodologies regarding the problem and the data. Since, this is a regression problem the safest approach was to create a simple MLP (Multi-layer Perceptron) and after that adding extra layers to make it more accurate. This way is time consuming but efficient, since every detail can contribute to achieve high performance.

Model architecture has 5 Dense layers (or 16 total layers):

- 1) First input dense layer:
 - a. Units: 256
 - b. kernel_initializer: he_normal
 - c. Input: number of features
 - d. BatchNormilization
 - e. Activation function (Relu)
 - f. Dropout (20%)
- 2) Hidden layers:
 - a. Dense layer with 256 units (same layers as before)
 - b. Dense layer with 128 units (same layers as before but Dropout change to 10%)
 - c. Dense layer with 64 units (same layers as before except no dropout)
- 3) Output layer:
 - a. Dense layer with 1 unit he_normal kernel_initialize and linear activation function

This descending approach will help the model to understand more difficult patterns. Adding also many dropout layers will help and stabilize the model performance to escape overfitting.

Also, adding optimizer help the model to make more stable steps and have better performance. Starting with a very low learning rate (0.0001) and after increase the value if performance is low.

Before adding all data into the model, it is important to mention that in XGBoost trained with non-normalized data, since it is not needed. In this problem it is essential to normalize in the same scale to

prevent model give more importance to some features than the others. Binary and one hot encoding labels are not scalable and so there is a list with only scaled columns:

- age
- utl.exp.ppp17
- hsize
- num.children5
- num.children10
- num.children18
- num.adult.female
- num.adult.male
- num.elderly
- sworkersh
- share.secondary
- Sfworkersh

Standard Scaler was selected to scale these data and labels values and weights so everything can be in the same scale preventing scaling unnecessary features.

Model was trained with the following parameters:

- Early Stopping (metric = val_mean_absolute_error, patience = 15)
- ReduceLROnPlateau (metric=val_mean_absolute_error, factor=0.2, patience=5, min_lr = 0.00001)
- Epoch = 100
- Batch_size = 32

After training model had better results than the other models (Figure 7). Results might mismatch with previous since now data are scaled. Starting with the loss value there is smooth decreasing from 0.52 - > 0.29. Val loss dropping rapidly to 0.3 which stay a bit stable. This is a result of the optimizer. Train and validation loss has small differences (0.29, 0.30) which means there is no overfitting, model can generalize well, and the model is stable.

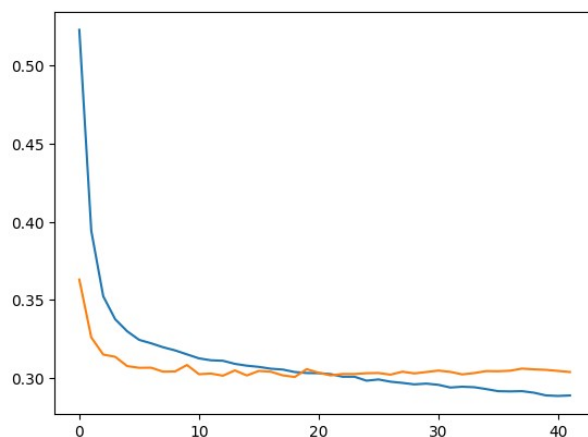


Figure 9: Deep learning model training plots
(blue line: test set, orange line: train set)

Since metrics are in different scale it is not possible to provide a clear answer if MLP is better or worse than XGBoost. The final answer in this question is coming from the challenge results in which MLP achieve a better position (Figure 10). The difference is a bit high which means that XGBoost is not so stable to generalize better into the new data rather than MLP.